

BAST: Blockchain-Assisted Secure and Traceable Data Sharing Scheme for Vehicular Networks

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Abstract—In vehicular networks, caching service content on edge servers (ESs) is a widely accepted strategy for promptly responding to vehicle requests, reducing communication overhead, and improving service experience. However, implementing such an architecture requires addressing the challenges associated with ES response data reliability and communication security. In this study, to tackle the ES response data reliability issue, a blockchain-assisted threshold signature scheme for cache-based vehicular networks is proposed. The scheme utilizes a threshold mechanism to sign the data broadcast by the ES, incorporates blockchain to trace malicious signers, and avoids the shortcomings and limitations associated with idealized assumptions for the ES in existing data-sharing schemes. Moreover, considering the communication security and high-speed mobility of vehicles, using the non-interactive signatures of knowledge based on the Σ -protocol, a secure and efficient message authentication scheme for vehicles and ESs is provided. Through rigorous security proofs and comprehensive analyses, our scheme satisfies the communication security requirements of vehicular networks. By leveraging the JPBC library for performance analysis, the proposed scheme demonstrates advantages as concerns both computation and communication overheads compared to related schemes. Moreover, we implemented the proposed scheme on an Ethereum test network (i.e., Goerli) to validate its feasibility.

Index Terms—Vehicular networks, blockchain, threshold signature, signatures of knowledge, authentication.

I. INTRODUCTION

THE rapid advancement of information technology has paved the way for a new paradigm of service-oriented

applications in vehicular networks. In the vehicular networks, vehicles within the network download application services, encompassing navigation maps, voice broadcasts, and entertainment content from the application service provider (ASP) to improve the driving experience [1]. However, accessing data directly from the ASP poses significant challenges for latency-sensitive applications and computationally intensive tasks in vehicular networks. Employing this method not only engenders significant consumption of broadband resources, placing substantial pressure on the network infrastructure, but also leads to considerable delays in the communication process [2]. Moreover, when vehicles submit a large volume of service requests, this can overwhelm the ASP and result in its performance bottleneck [3].

To address the limitations associated with vehicles obtaining service data directly from the ASP, researchers have proposed caching the ASP service content to an edge server (ES) in close proximity to the vehicle [4]. Therefore, vehicles can directly access service content from the ES, mitigate data center bandwidth pressure, and enhance the driving experience [5]. However, in most cache-based vehicular networks, messages transmitted between vehicles and ESs or roadside units (RSUs) often utilize open and insecure communication channels, making them vulnerable to interception, modification, and replay attacks [6]. To address these issues, an authentication protocol is needed to ensure the security of message transmission in cache-based vehicular networks [7]. A well-designed authentication protocol should possess essential properties such as authentication, integrity, and non-repudiation. Together, these properties ensure that authentication schemes in cache-based vehicular networks can effectively protect user privacy and communication security [8] and [9].

Most existing authentication schemes for vehicular networks depend on a centralized and fully trusted remote authentication server to manage registration, authentication, and key distribution for vehicles and other entities (i.e., RSUs and ESs) [10]. The centralized authentication schemes in vehicular networks require frequent interactions between the remote authentication server, vehicles, and other entities. However, these multi-party interactions introduce significant authentication and communication delays, making them unacceptable for resource-constrained vehicles (e.g., in vehicle-to-vehicle scenarios based on IEEE 802.11n devices, the maximum coverage range is 850 meters, and the throughput is 15 Mbps, in vehicle-to-infrastructure scenarios based on IEEE 802.11n, when the vehicle passes through the three infrastructures at a speed of 100 km/h, the average effective throughput is

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9.83 Mbps) [11], [12]. Furthermore, centralized schemes are vulnerable to single-point attacks, causing the leakage of user privacy data (e.g., the attackers can use side channel attacks to compromise the system and obtain the system private key) [13]. Clearly, constructing a remote authentication server that fulfills these ideal security requirements is costly.

Blockchain provides a new approach to addressing the limitations of centralized vehicular networks authentication schemes [14], [15]. By abstracting data sharing between vehicles into a transaction process between nodes in the blockchain, and using a consensus algorithm to verify and record these transactions, the shared data is recorded in the blockchain and encrypted using hash encryption [16]. This process ensures the tamper-proof and privacy of the data and can meet the requirements for communication security and data integrity in vehicular networks [17]. The chain structure and timestamping ensure the continuity of vehicle data-sharing transactions and the traceability of data sharing. Additionally, the anonymity of the blockchain, combined with a public-key-based authentication mechanism, effectively protects user privacy. Through blockchain, vehicular networks can achieve a more secure and reliable method of data sharing.

While blockchain-based vehicular networks enhance security and enable distributed data storage for vehicle data sharing, they also have certain limitations [18]. Specifically, in vehicular networks, RSUs and base stations (BSs) function as nodes within the blockchain, assuming critical roles in maintaining network integrity and processing transactions. These infrastructures connect with vehicles through communication technologies (e.g., IEEE 802.11p or cellular communication network), enabling vehicles to request and receive services by executing blockchain transactions. In certain scenarios, vehicles upload shared data to the blockchain, enabling drivers and passengers to stay informed about road conditions [19]. However, malicious vehicles may spread false or harmful data, causing traffic disruptions. The traceability of blockchain provides favorable conditions for tracing malicious vehicles, enhancing the overall security and reliability of vehicular networks. Additionally, while vehicles have limited computing resources, they may still need to employ public key or symmetric encryption algorithms to ensure secure communication and protect the privacy of shared data [20]. In light of these challenges, the following questions naturally arise:

“How can traceability and communication security requirements be met with acceptable vehicle performance overhead?”

A. Contributions

In this study, we propose a blockchain-assisted secure and traceable data sharing scheme for cache-based vehicular networks (BAST), which gives a positive answer to the above questions. The main contributions of this work are as follows.

- 1) We propose a blockchain-based threshold signature scheme that employs threshold signatures for data broadcast by ES and utilizes RSUs to generate aggregate

signatures. By integrating blockchain into the cache-based vehicular networks system, this scheme mitigates the need for complete trust in the ES and provides a traceable function for malicious signers. This design makes our proposed signature scheme be more suitable for the application of response data reliability in cache-based vehicular networks data sharing.

- 2) We propose a non-interactive knowledge signature scheme based on the Σ -protocol. Notably, the proposed scheme eliminates the need for a remote authentication server, supporting efficient and fast authentication between vehicles and ESs. The design can make our proposed signature scheme be more suitable for securing delay-sensitive applications in cache-based vehicular networks data sharing.
- 3) The security analysis demonstrates that our scheme fulfills the security and privacy requirements of vehicular networks while providing additional functions. Through performance evaluation comparing our scheme with related schemes, the results show that the performance of the scheme is acceptable and suitable for deployment in vehicular networks.

B. Organization of the Rest Paper

In Sections II and III, the technology relevant to this study is introduced and some representative related works are reviewed. In Section IV, we provide the detailed construction of the proposed scheme. The security and performance analysis of this scheme are described in Sections V and VI. In Section VII, we provide a comprehensive summary of this scheme and indicate potential avenues for future research directions.

II. RELATED WORK

This section summarizes the research on privacy and security in vehicular networks, focusing on the aggregate signature protocol for cache-based vehicular networks and blockchain-assisted data sharing for vehicular networks.

A. The Aggregate Signature Authentication Protocol for Cache-Based Vehicular Networks

In [10], to ensure the secure data sharing for cache-based vehicular networks and establish secure communication between vehicles and fog nodes (FNs). Zhang et al. designed an efficient authentication scheme for vehicles and FNs cache content based on the TESLA protocol. Moreover, addressing the high-speed mobility of vehicles, a privacy-preserving challenge and response mechanism based on Pedersen commitment has been devised for communication between vehicles and FNs. However, Cheng et al. [7] pointed out that this scheme does not achieve message reliability.

In [3], through delegating authentication capabilities to distributed edge nodes (ENs), Yang et al. introduced an efficient and rapid mutual authentication protocol based on threshold cryptography. With the help of multiple ENs that collaborate, this scheme verifies the authenticity of the vehicle identity and issues tokens to permitted vehicles, which the vehicles use to

accelerate up the second verification phase. But, the scheme has high computation and communication overheads [21], and lacks considerations of access control to EN.

Later, to achieve access control for ENs in the cache-based vehicular networks, Zhang et al. [22] proposed a secure and efficient data sharing mechanism based on multi-authority ciphertext-policy attribute-based encryption. The scheme provides a fine-grained access control for ES, ensuring that only ES that complies with the vehicle requested access policy has the ability to provide vehicle the necessary service. Furthermore, this scheme facilitates efficient and fast authentication for vehicles and ESs by integrating token mechanisms. However, the attribute set has a significant effect on the computational overhead of this scheme.

Previous research has indicated that cache-based vehicular networks constitute a promising area of investigation. However, the openness of the vehicular networks communication environment presents significant challenges for the security communications. Moreover, [3], [10], and [22] inadequately addresses prevailing security and performance issues, such as ensure the reliability of messages received by the vehicle, and handle data simultaneously transmitted by multiple ESs. Fortunately, several schemes have been proposed to tackle these challenges in vehicular networks.

In [23], through employing message recovery signature technology, Shen et al. proposed a secure and efficient real-time traffic data aggregation scheme for vehicular networks. In this study, it achieves the validity verification of vehicle signatures, ensuring the security of traffic data. Moreover, it provides batch verification of signatures for multiple vehicles. In this research, the aggregator is fully trusted and is utilized to recover the original traffic data of vehicle [24], which makes this scheme unsuitable for cache-based vehicular networks.

In [25], Yang et al. proposed a privacy-preserving aggregate authentication scheme for the security warning system in fog-cloud vehicular networks, which employs pseudonym mechanism achieves the anonymity of the vehicle. Moreover, the scheme employs signcryption algorithms to achieve ciphertext aggregation and batch verification. However, the solution requires vehicles to be pre-loaded with pseudonyms, resulting in high storage burden, and vulnerable to collusion attacks [26].

For resist collusion attacks and establish traceability in vehicular networks, Zhou et al. [27] introduced an anonymous and traceable multi-authority ciphertext-policy attribute-based encryption scheme designed for secure communication in edge computing environments, which realizes access control to ENs. This scheme employs signatures of knowledge to trace malicious behaviors and provides detailed security proofs. However, the scheme entails high computational overhead and is unsuitable for vehicular networks characterized by high delay sensitivity.

According to the foregoing, while the aggregated signature authentication protocol holds considerable promise for applications in vehicular networks, it is burdened by unreasonable security assumptions, vulnerability to various attacks, and lacks the capability of trace malicious signers.

B. The Blockchain-Assisted Data Sharing Scheme for Vehicular Networks

In [28], a mutual authentication mechanism for Internet of Things (IoT) nodes was proposed, which utilizes both public and private blockchains to enhance the efficiency of node identity authentication and reduce dependence on remote authentication servers. Moreover, considering the requirements of delay-sensitive applications in the IoT environment, Cheng et al. [29] integrated elliptic curve cryptography (ECC) and blockchain to propose a mutual authentication scheme specifically designed for IoT devices. However, Wang et al. [30] pointed out that the authentication process of this scheme incurs a high computational overhead.

In [31], to mitigate computational overhead and enhance data sharing efficiency, Wang et al. introduced a blockchain-assisted lightweight cross-domain message authentication scheme for the IoT. This scheme employs ECC and hash functions to realize lightweight cross-domain communication for ES-assisted smart devices. However, this scheme involves numerous interactions, and the ES assumptions are idealized.

Although the integration of blockchain into IoT for secure data sharing holds significant research potential, the direct application of blockchain in vehicular networks faces major challenges because to the frequent changes in network connections and the high mobility of vehicle. Fortunately, researchers have devised innovative scheme to tackle these issues.

To enhance the security of vehicular networks communication and improve the overall driving experience, Feng et al. [32] proposed a privacy-preserving scheme for vehicular networks by employing blockchain and attribute-based encryption, which eliminates the need for a remote authentication server to verify the credibility of transmitted messages. Moreover, Lin et al. [33] employed signatures of knowledge and smart contract to design a traceable one-time public key authentication scheme. However, these schemes may be vulnerable to compromised authorities attacks [34].

In [35], to address various attacks and ensure the reliability of vehicle broadcast messages, Li et al. employed threshold signatures and smart contracts to design a secure and trustworthy announcement dissemination scheme for vehicular cloud networks. Specifically, with the assistance of nearby vehicles, this scheme utilizes threshold signatures and the reputation mechanism of the blockchain to ensure the reliability of the messages that vehicles broadcast. However, this scheme neglects the utilization of ENs to broadcast traffic information, aiming to alleviate the computation overhead on vehicles.

To fulfill the demands for frequent authentication of vehicles and ENs in vehicular networks, while ensuring a seamless service experience, Shen et al. [6] presented a secure and efficient blockchain-assisted authentication scheme designed for ENs-integrated vehicular networks. In this scheme, the EN directly queries the authentication results in the consortium blockchain, enabling swift authentication of vehicles and edge nodes. However, this scheme does not consider collusion attacks and lacks a trace mechanism for malicious ENs.

Based on the aforementioned survey, while integrating blockchain into cache-based vehicular networks holds

promising research prospects, the establishment of a reliable and traceable data sharing scheme presents considerable challenges. Moreover, the imperative to design an efficient and fast authentication scheme based on vehicle characteristics is crucial in cache-based vehicular networks.

III. BACKGROUND

A. System Model and Assumptions

Fig. 1 shows the cache-based data sharing system model in vehicular networks, the trust authority (TA), blockchain, ES, RSU, and vehicle are among the parties involved in this study. The main communication methods are vehicle/RSU/ES/-to-TA (V2T), vehicle-to-vehicle (V2V), vehicle-to-RSU/ES (V2I), and vehicle-to-blockchain (V2B). V2T is for vehicle/RSU/ES/ to obtain encryption materials from TA through offline registration. V2V is for vehicle-to-vehicle communication using the IEEE 802.11p protocol. V2I is for vehicles to communicate with RSUs/ESs to obtain traffic information (e.g., road conditions). V2B is for vehicles to request blockchain services to trace malicious message signers or share ES traffic status. The main functions and security assumptions of each entity are described as follows.

- **TA:** It serves the responsibilities of the creating system parameters. Moreover, it is responsible for generating corresponding encryption materials for entities, deploying smart contracts, and facilitating secure communication.
- **Vehicle:** Each vehicle has an onboard device that securely and dependably takes out encryption and decryption processes. Moreover, it is employed to confirm that the requested message is correct and that the aggregate signature is legitimate.
- **ES:** It is installed on both sides of the road, where it uses the ASP to cache application and traffic-related data. In response to vehicle queries, the ES is in charge of disseminating the service type list and delivering related information. Moreover, to enhance the credibility of the message, multiple ESs independently sign the message.
- **RSU:** It serves as a wireless infrastructure positioned alongside roads, primarily tasked with providing network connectivity, forwarding service information, and various services to vehicles. Each RSU controls several ESs and is in charge of aggregating the signatures that the ESs produce.
- **Blockchain:** The blockchain provides immutable, undeniable, and verifiable data storage, which facilitates the formation of transactions and the construction of blockchains [36]. Specifically, we embed traceable private keys within the transaction, enabling the identification of malicious signers based on these private keys, rather than requiring the pre-loading of all traceable private keys in the TA. We recommend that the BS (i.e., BS is assumed to be a fully trusted entity and does not provide a pseudo API [33]) join the network as a full node to provide services to nearby RSUs and vehicles.

In this study, we assume that the onboard unit sensitive material is uncompromised, and consider the TA as a completely trustworthy entity it cannot be cracked [10]. To

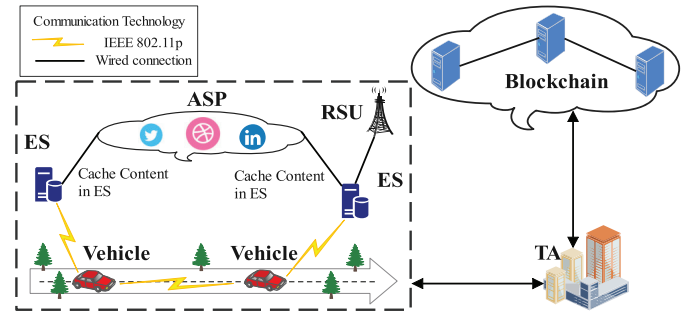


Fig. 1. The cache-based vehicular networks data sharing system model.

get closer to real, we assume RSU is honest but curious participants [22].

B. Security Objectives

To better reflect real-world scenarios, we assume that ES may be malicious or compromised by attackers. However, to ensure system security, we also assume that the attacker can compromise only a limited number of ES, with this limit determined by specific system parameters. The communication channel between vehicles and ESs/RSUs is assumed to be vulnerable to attackers who may eavesdrop, inject, transmit, or even modify messages. Therefore, ensuring security and privacy is paramount for maintaining secure communication between vehicles and ESs/RSUs. In addition, we need to consider the impact of blockchain with consensus mechanisms on security objectives. We conducted a survey of existing vehicular networks focusing on cached aggregate signature authentication protocols such as [3], [10], [25], and [26], and blockchain-based data sharing schemes such as [6], [32], [35], and [37]. Based on this survey, we propose the following security requirements for the BAST protocol.

- **Message authentication:** In vehicular networks, it is imperative to authenticate the authenticity and legality of transmitted messages. In this study, any altered or incorrect message cannot successfully pass the authentication process.
- **Malicious ES traceability:** In real-world scenarios, ES is assumed to be malicious or easily cracked by an attacker. If an adversary cracks ES to obtain the signature private key, the attacker/malicious ES may use the private key to sign a false message and broadcast the message. Therefore, it is necessary to make malicious ES traceable.
- **Resistance to 51% attacks:** In blockchain-based schemes, the attacker can manipulate individual nodes to verify fake blocks they generate by forcing them to cease confirming legitimate blocks [38]. Therefore, the BAST protocol must ensure that an attacker cannot control the majority of computing or consensus power (e.g., hashrate in PoW [33]). If such control is achieved, the attacker could reverse and modify historical transactions to obtain trace keys and disseminate false messages.
- **Resistance collusion attacks:** Malicious ESs may collude with RSUs to generate fraudulent aggregate signatures and tamper with transmitted messages, leading

to the broadcast of false information. Additionally, an attacker might control a group of ESs with the intent of compromising the message signing mechanism, allowing any message (including malicious or false information) to be broadcast irrespective of its legitimacy. Therefore, the scheme must resist collusion attacks and mitigate security risks to the system.

- **Resistance to modification attacks:** If the message obtained by the receiver is inconsistent with the signed message, the receiver must reject the message because the message may have been modified and broadcast by an attacker.
- **Resistance to impersonation attacks of vehicle:** The attacker may use the identity information of the legitimate vehicle to request the blockchain to trace the malicious ES, then the blockchain should reject the message.
- **Resistance to replay attacks:** A malicious sender is unable to collect and store a signed message to deliver it after the original message has expired, making the receiver believe that the communication is legitimate.

C. Schnorr Signature

In this study, the primary steps of the Schnorr signature algorithm and hardness assumptions are as follows [39].

KeyGen(λ) $\rightarrow$ (sk, pk). Enter implicit security parameters λ in the system. It randomly selects $sk \in \mathbb{Z}_p^*$ as the signing key, and computes the corresponding public key $pk = g^{sk}$.

Sign(sk, m) $\rightarrow \sigma$. The algorithm inputs the signing key sk and the message m . The process of generating a signature σ is as follows. It randomly selects an $r \in \mathbb{Z}_p^*$, computes the public key $R = g^r$, sign the R , m , and generates $c = H(pk, R, m)$, $z = r + sk \cdot c$. Then, it outputs the signature $\sigma = (R, z)$.

Verify(pk, m, σ) $\rightarrow$ 0/1. The algorithm takes the pk , m , σ as input, and the signature verification process is as follows. If the $c = H(pk, R, m)$ and $g^z = pk^c R$ indicate success output 1, otherwise, output 0.

- **Discrete logarithm assumption:** The discrete logarithm assumption (DL) is defined as follows. Assume that give $g, \alpha \in \mathbb{G}$, to compute $\beta \in \mathbb{Z}_p^*$ such that $\alpha = g^\beta$ is negligible. The following is the definition of the adversary $\mathcal{A}$ that can solve the DL.

$$\Pr[(\mathbb{G}, g, p) \leftarrow K(n), \beta \in \mathbb{Z}_p^* : \alpha \leftarrow \mathcal{A}(\mathbb{G}, g, p, g^\beta)] \leq \text{negl}(n).$$

D. ElGamal Encryption

The Diffie-Hellman computational assumption is the foundation of the ElGamal encryption algorithm. According to [27], the following outlines the specific steps of the algorithm.

KeyGen(1^n) $\rightarrow$ (sk, pk). Enter implicit security parameters n in the system. It randomly chooses secret keys $sk \in \mathbb{Z}_p^*$, and calculates the corresponding public key $pk = g^{sk}$.

Enc(pk, m) $\rightarrow ct$. In the encryption algorithm, input the pk and the message to be encrypted m . It randomly chooses $u \in \mathbb{Z}_p^*$, and computes encrypted messages $ct_1 = g^u$, $ct_2 = mpk^u$. Then, it outputs the ciphertext $ct = \{ct_1, ct_2\}$.

Dec(sk, ct) $\rightarrow m$. The decryption process involves input the secret keys sk , ciphertext ct , and the message m can be decrypted by calculating $m = \frac{ct_2}{ct_1^k}$.

E. Signatures of Knowledge

The signature of knowledge represents a distinctive signature technology that facilitates the process of signing messages in a zero-knowledge manner [40]. This innovative approach allows the signer to provide a signature without disclosing any information or knowledge surrounding the signed message. The specific steps of the algorithm are summarized below.

Setup(1^k) $\rightarrow pp$. Enter implicit security parameters k in the system. Then, the system outputs public parameters pp .

Sign(pp, m, x, w) $\rightarrow \sigma$. The algorithm inputs the pp , a message m , and a difficulty relation $(x, w) \in R$ (where w is a valid witness for x , and x is a statement). Then, it outputs the signature σ .

Verify(pp, m, x, σ) $\rightarrow$ 0/1. The algorithm inputs the pp , m , x , and the signature σ . Then, if the σ is valid outputs 1, otherwise, it outputs 0.

F. Non-Interactive Zero Knowledge Argument

The non-interactive zero-knowledge argument, encompasses three key components (i.e., a single message that was exchanged between prover $\mathcal{P}$ and verifier $\mathcal{V}$). Without learning any additional relevant knowledge, the verifier can convince the prover that they have certain values. Let R be a relationship that defines a language within NP, and $(x, w) \in R$, where w is a witness. It consists of three algorithms (i.e., $\mathcal{K}$, $\mathcal{P}$, $\mathcal{V}$) and satisfies the following properties [41].

- **Completeness:** For each $\mathcal{K}(1^n) \rightarrow crs$ and $(x, w) \in R$. If the probability equals 1, property is considered perfect. This property is specifically described as follows.
 $\Pr[\pi \leftarrow \mathcal{P}(x, w, crs) : \mathcal{V}(x, \pi, crs) = 1] \geq 1 - \text{negl}(n)$
- **Soundness:** For all probabilistic polynomial time (PPT) adversary $\mathcal{A}$, the probability as follows.

$$\Pr \left[\begin{array}{l} crs \leftarrow \mathcal{K}(1^n), (x, \pi) \leftarrow \mathcal{A}(crs) : \\ x \notin R \wedge \mathcal{V}(x, \pi, crs) = 1 \end{array} \right] \leq \text{negl}(n).$$

- **Zero-knowledge:** The proof sent by $\mathcal{P}$ to $\mathcal{V}$ does not disclose any secret or additional information to $\mathcal{V}$. Therefore, the $\mathcal{V}$ gains no knowledge of secret information from the proof.

G. Σ -Protocol

According to [42], the Σ -protocol is regarded as a 3-move process that operates on the relationship R , where $(x, w) \in R$ is a computable binary relation, x is a statement, and w is a witness. The Σ -protocol is executed by a prover $\mathcal{P}$ and a verifier $\mathcal{V}$ that receive a common input x , finish interaction, a triple (a, c, z) will be generated as the output of the protocol. It has the following properties and hardness assumptions.

- **Completeness:** If $\mathcal{P}$ and $\mathcal{V}$ execute the protocol, input $(x, w) \in R$ is always received by $\mathcal{V}$.
- **Special soundness:** It is able to construct witness w for each statement x employing the PPT algorithm *Ext*, given two accepted (x, a, c, z) and (x, a, c^*, z^*) , where $c \neq c^*$.

- **Special honest verifier zero knowledge:** From a known c , there is a PPT algorithm Sim that produces an acceptable (x, a, c, z) that is identical to the result of the actual interaction between honest $\mathcal{P}$ and $\mathcal{V}$.
- **Decisional diffie-hellman assumption:** The decisional diffie-hellman assumption (DDH) is defined as follows. Assume that give $g, g^\alpha, g^\beta, g^r \in \mathbb{G}$, decide $r = \alpha\beta$ or $r \in \mathbb{Z}_p^*$ is negligible. The $\mathcal{A}$ can solve the DDH is defined as follows.

$$\left| \Pr \left[\begin{array}{l} (\mathbb{G}, p, g) \leftarrow \mathcal{K}(1^n), \alpha, \beta, r \in \mathbb{Z}_p^* : \\ \mathcal{A}((\mathbb{G}, p, g), g^\alpha, g^\beta, g^r) = 1 \end{array} \right] - \Pr \left[\begin{array}{l} (\mathbb{G}, p, g) \leftarrow \mathcal{K}(1^n), \alpha, \beta, r \in \mathbb{Z}_p^* : \\ \mathcal{A}((\mathbb{G}, p, g), g^\alpha, g^\beta, g^{\alpha\beta}) = 1 \end{array} \right] \right| \leq \text{negl}(n).$$

H. Security Definition

• **Definition 1 (Existential Unforgeability Against a Chosen Message Attack with Traceability):** In this scheme, while meeting the requirements of unforgeability against a chosen message attack, also needs to meet the requirements of accountability. According to [43], the game defines the advantage that the adversary has over the security parameter λ in terms of unforgeability and traceability against the BAST scheme. The adversary obtains the signature share and the signer's private key for the generated message m from the set C . However, the adversary cannot generate a valid signature σ using the acquired information, preventing the *Trace* from either failing or incorrectly identifying the signer outside of set C . This condition reflects the property of traceability. If the signature share obtained by the adversary is below the threshold t , a valid signature σ can still be generated. In this phase, the *Trace* assigns some responsibility to the subset C_t . According to the definition of the *Trace* in BAST, when $|C_t| \geq t$, C_t is excluded from participating in the traceability process. Therefore, the adversary has successfully compromised the honest party and won the game. If the adversary fails to win this game, BAST is considered unforgeable.

IV. PRESENTATION OF BAST

A. The Algorithm Description of the BAST

In the traditional threshold signature scheme, signatures contributed by multiple users are aggregated, with an aggregator responsible for generating aggregate signatures [44]. However, there may be illegal signers signing messages, and the aggregator may aggregate the signature information of malicious signers without being aware of their illegitimacy. In this study, there exists a tracker that has the ability to trace the identity of an illegal signer if it possesses the corresponding trace key. According to [45], the following is an algorithm description of the proposed scheme.

KeyGen(λ, n, t) $\rightarrow (pk, sk, sk_{ag}, sk_t)$. In the execution of the **KeyGen** algorithm, the security parameter λ , the number of parties n , and the threshold t are all inputted as parameters. Then, it outputs public key pk , private key sk of the ES, the sk_{ag} for the RSU, and sk_t for the blockchain.

Sign(sk_i, m, C) $\rightarrow \sigma_i$. The inputs for the **Sign** algorithm are the signer who uses its secret key sk_i , data m , and C , where

C represents the number of participating signers ES. Then, it generates the corresponding signature σ_i for the ES.

Aggregate($sk_{ag}, m, C, \{\sigma_i\}_{i \in C}$) $\rightarrow \sigma$. The inputs for the **Aggregate** algorithm are the sk_{ag} , data m , C , and the signature information σ_i generated from different ESs. Then, it generates aggregate signature σ and sends the vehicle.

Verify(pk, m, σ) $\rightarrow 0/1$. The **Verify** algorithm employs the input public key pk to verify the legitimacy of the signature σ on message m . If σ is a valid signature, it outputs 1, otherwise, it outputs 0. **Trace**(sk_t, m, σ) $\rightarrow C/\text{fail}$. The blockchain employs a **Trace** algorithm to trace the members of the signature based on the input trace key sk_t , the σ , and m . If the signature member is successfully traced, the algorithm outputs the member set C . Otherwise, it outputs fail.

B. Detailed Construction of the BAST

The BAST framework is illustrated in Fig. 2 and can be summarized as follows. First, during the system initialization phase, the TA sets the system parameters and provides registration services for ES, RSU, and vehicle. Then, multiple ESs collaboratively sign the broadcast message, which is then sent to the RSU for the generation of an aggregate signature. The RSU forwards the signature and message to the vehicle, which verifies the message and checks its legitimacy. If the verification fails, the vehicle submits a malicious signature trace request to the blockchain. Finally, leveraging the immutability of blockchain to trace the malicious signer.

• **System Initialization:** During this process, TA initializes the system parameters and generates corresponding public, private keys and other encryption materials for ES, RSU, and vehicle, respectively.

KeyGen(λ, n, t) $\rightarrow (pk, sk, sk_{ag}, sk_t)$. In the phase, the TA selects a prime order p group $\mathbb{G}_1$ with two independent generators g, h , and a hash function $H : \{0, 1\}^* \rightarrow \mathbb{Z}_p^*$. Subsequently, computes the public and private key pair of ES according to n , randomly selected $sk_1, \dots, sk_n \in \mathbb{Z}_p^*$ (i.e., $sk = \{sk_1, \dots, sk_n\}$), calculates the corresponding public key $pk_i = g^{sk_i}, i \in [1, \dots, n]$ (i.e., $pk^* = \{pk_1, \dots, pk_n\}$). The TA uses ElGamal algorithm to encrypt t , chooses randomly $\omega \in \mathbb{Z}_p^*$, computes $T_0 = g^\omega$, and $T_1 = g^t h^\omega$. Furthermore, the TA provides the randomly generated private keys $sk_e, sk_{rs} \in \mathbb{Z}_p^*$, and computes the corresponding public keys $pk_e = g^{sk_e}$ and $pk_{rs} = g^{sk_{rs}}$, respectively. Here, TA generates a pair of (sk_{TA}, vk_{TA}) for sign and verify, where vk_{TA} is known publicly. During this period, TA assigns a unique identity vid_i for each legal vehicle, randomly generates the private key $k_{vid_i} \in \mathbb{Z}_p^*$, and calculates the corresponding public key $pk_{vid_i} = g^{k_{vid_i}}$. Moreover, TA generate a unique certificate $Cert_{vid_i}$ for the vehicle for identity verification.

Finally, the TA sends $pk = \{pk^*, pk_t, pk_{rs}, vk_{TA}, T_0, T_1\}$ and $\{vid_i, k_{vid_i}, Cert_{vid_i}\}$ to the vehicle for verification. Meanwhile, the TA provides the combined key $sk_{ag} = \{pk, sk_{rs}, t, \omega\}$ and the trace key $sk_t = \{pk, sk_e, t\}$ to the RSU and blockchain, respectively. Moreover, each ES obtains its own signature secret key sk_i issued by TA.

• **ES Generate Signature:** **Sign**(sk_i, m, C) $\rightarrow \sigma_i$. In the phase, to enhance the reliability of message m , multiple ESs

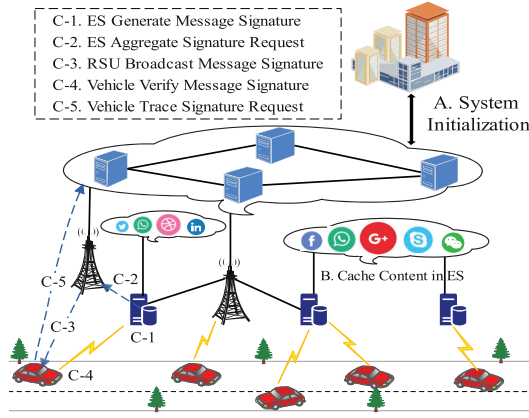


Fig. 2. Framework of the BAST scheme.

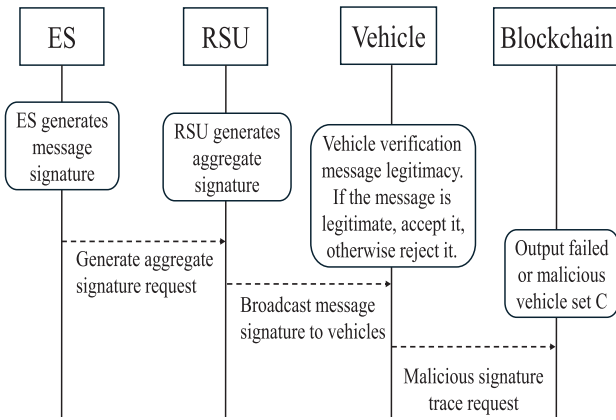


Fig. 3. Sequence diagram of the BAST scheme.

collaborate to sign the broadcasted message m from the head ES. The ES selects randomly $r_i \in \mathbb{Z}_p^*$, computes $R_i = g^{r_i}$, $c_i = H(pk, R_i, m)$, and $z_i = r_i + sk_i \cdot c_i$. Finally, the ES sends the signature $\sigma_i = (R_i, z_i)$ and message m to the RSU for generate the aggregate signature. Moreover, to make it easier to understand the interaction between entities, we give a sequence diagram (i.e., Fig. 3).

• **RSU Aggregate Signature:** $Agg(sk_{ag}, m, C, \{\sigma_i\}_{i \in C}) \rightarrow \sigma$. In the phase, the RSU receives the signature information σ_i from the ES. Subsequently, RSU obtains R_i with each ES taking part in the present signing procedure, computes $R = \prod_{i \in C} R_i$, $z = \sum_{i \in C} z_i$, $c = H(pk, R, m)$, and checks $g^z = [\prod_{i \in C} pk_i] \cdot R$ through the signature information obtained by ES. Next, RSU uses the ElGamal algorithm to encrypt z , chooses randomly $\rho \in \mathbb{Z}_p^*$, computes $ct_1 = g^\rho$, $ct_2 = g^z \cdot pk_\rho^\rho$. The RSU chooses randomly $(b_1, \dots, b_n) \in \{0, 1\}^n$, if $i = C$, calculates the $g^z = [\prod_{i=1}^t (pk_i)^{b_i}] \cdot R$, where $b_i = 1$, C is a set of size t of participating signers. Then, RSU generates knowledge proof $\pi = \{z, \rho, \eta, w, \kappa_1, \dots, \kappa_n, \varepsilon_1, \dots, \varepsilon_n \in \mathbb{Z}_p^*\}$ that can prove that R and the encrypted z are valid. The RSU uses private key sk_{rs} to sign the $(sk_{rs}, (m, R, ct_1, ct_2, \psi))$ for generate tg . Finally, the RSU sends $\sigma = \{R, ct_1, ct_2, \pi, tg, TS_{exp}\}$ and m to the vehicle, where TS_{exp} is a timestamp.

• **Vehicle Verify Signature:** $Verify(pk, m, \sigma) \rightarrow 0/1$. In this phase, once the vehicle receives the signature σ sent by

RSU, the vehicle performs the following steps. To ascertain whether the message is new, the vehicle evaluates whether the equation $TS_c \leq TS_{exp}$ holds, where TS_c represents the timestamp at when the message was received. If not, vehicle drops off the message. Otherwise, the vehicle verifies the legitimacy of signature and consistency of required message.

Due to the $pk = \{pk^*, pk_t, pk_{rs}, T_0, T_1\}$ and computes $c = H(pk, R, m)$. If the vehicle uses the Schnorr verification algorithm accepts the $(pk_{rs}, (m, R, ct, \pi), tg)$, i.e., $Verify(pk_{rs}, (m, R, ct, \pi), tg) = 1$. Moreover, the following is the calculation of vehicle.

$$R \cdot \left[\prod_{i=1}^t pk_i^{k_i} \right]^c \stackrel{?}{=} g^z, \quad ct_1 \stackrel{?}{=} g^\rho, \quad ct_2 \stackrel{?}{=} pk_t^\rho \cdot g^z.$$

$$T_0 \stackrel{?}{=} g^\lambda, \quad T_1 \stackrel{?}{=} \left(\prod_{i=1}^n g^{k_i} \right) \cdot h^\lambda, \quad v_0 \stackrel{?}{=} g^\eta, \quad v_i \stackrel{?}{=} g^{k_i} h_i^\eta,$$

$$\prod_{i=1}^n v_i^{a_i} \stackrel{?}{=} \prod_{i=1}^n h_i^{v_i}, b_i(1 - b_i) \stackrel{?}{=} 0, (i.e., b_i \in \{0, 1\}).$$

Algorithm 1 Blockchain Trace Malicious Signature

Input : The vehicle sends $\{vid_i, Cert_{vid_i}, m, \sigma, TS_{vid_i}\}$ to blockchain

Output: Malicious signer C / fail.

```

1 Step 1: Vehicle identity verification
2 if  $Verify_{vk_{TA}}(Cert_{vid_i}, vid_i) \rightarrow true$  then
3   if  $TS_B \leq TS_{vid_i}$  then
4     return an error.
5   end
6   if  $Use\ Schnorr\ signature\ verify(pk, m, \sigma) \rightarrow false$  then
7     drop connection.
8   end
9 end
10 else if then
11   drop connection and return error.
12 end
13 Step 2: Blockchain trace malicious signers
14 if  $\sigma \rightarrow \{R, ct_1, ct_2, \pi, tg\} \rightarrow true$  then
15   computes  $c = H(pk, R, m)$ ,
16   use ElGamal algorithm decrypt  $\{ct_1, ct_2\}$ ,
17   find  $C \subseteq [n]$ , computes  $pk_C = \prod_{i \in C} pk_i$ ,
18   if  $g^{z^*} = R \cdot (\prod_{i \in C} pk_i)^c$  then
19     return  $C$  to the vehicle.
20   end
21 end
22 else if then
23   drop connection and return fail to the vehicle.
24 end

```

If the aforementioned equation is validated, it signifies the correctness of the message, allowing for its reception by the vehicle. Otherwise, in the event of a failed verification, the reception of the message will be declined, necessitating the reporting of such message to the blockchain to assist in the identification and trace of the signing member responsible for the message.

• **Blockchain Trace Signature:** $Trace(sk_t, m, \sigma) \rightarrow C/fail$. As shown in Algorithm 1, blockchain trace malicious signers mainly steps as follows. The blockchain obtains the service

request messages $\{\text{vid}_i, \text{Cert}_{\text{vid}_i}, m, \sigma, TS_{\text{vid}_i}\}$ from the vehicle. Then, the blockchain employs verification processes to validate the legitimacy of vehicles. Following a successful verification, by determining whether the equation $TS_B \leq TS_{\text{vid}_i}$ holds, where TS_B denote the blockchain receives message timestamp. The blockchain drops off the connection if the timestamp expired. Otherwise, to determine whether (m, σ) is a legitimate signature, the blockchain employs the Schnorr signature verification algorithm. If verify fail, blockchain returns error.

If the above verification is successful. The blockchain retrieves $\{R, ct_1, ct_2, \pi, tg\}$ based on vehicle request messages. Then, computes signature $c = H(pk, R, m)$, chooses randomly $z^* \in \mathbb{Z}_p^*$, uses the ElGamal algorithm decrypt $\{ct_1, ct_2\}$, and computes $g^{z^*} = \frac{ct_2}{ct_1^{sk_e}}$. Then, the blockchain finds $C \subseteq [n]$, calculates public key $pk_C = \prod_{i \in C} pk_i$, where the size of C is equal to t . The blockchain calculation and verification whether the equation $g^{z^*} = R \cdot (pk_C)^c$ holds. If not, the blockchain then drops the connection and returns failure to the vehicle. Otherwise, the blockchain sends C to the vehicle and records the signers in the set C in the blockchain.

V. SECURITY ANALYSIS

In this section, we perform security proofs for the Σ -protocol, utilizing both formal and informal proofs to analyze the security of the proposed scheme. Here, we reference the methods outlined in [27] and [33] to establish the security of the proposed scheme structure.

A. Σ -Protocol Proof

We elucidate the Σ -protocol through the interaction between the prover $\mathcal{P}$ and the verifier $\mathcal{V}$. Subsequently, provide the security proof that this scheme satisfies the random oracle model of knowledge signature security requirements. We provide the Σ -protocol statement of interaction as follows [46].

$$\begin{aligned} &\{g, h, h_1, \dots, h_n, pk_1 = g^{sk_1}, \dots, pk_n = g^{sk_n}, pk_t = g^{sk_e}, \\ &T_0 = g^\lambda, T_1 = g^t h^\lambda, R = \prod_{i \in C} R_i, ct = (ct_1, ct_2), v_0 = g^\eta, \\ &v_1 = g^{K_1} h_1^\eta, \dots, v_n = g^{K_n} h_n^\eta, \alpha, c = H(pk, R, m)\}, m \end{aligned}$$

Moreover, the $\mathcal{P}$ has the witness $z, \rho, \eta, w, \kappa_1, \dots, \kappa_n, \varepsilon_1, \dots, \varepsilon_n$, and selects $f_z, f_\rho, f_\eta, f_w, (f_{\kappa_1}, \dots, f_{\kappa_n}), (f_{\varepsilon_1}, \dots, f_{\varepsilon_n}) \in \mathbb{Z}_p^*$ randomly. The $\mathcal{P}$ computes $F_1 = g^{f_z} \cdot \prod_{i=1}^t pk_i^{c \cdot f_{\varepsilon_i}}, F_2 = g^{f_\rho}, F_3 = pk_t^{f_\rho} \cdot g^{f_z}, F_4 = g^{f_w}, F_{5i} = g^{\sum_{j=1}^n f_{\varepsilon_j} \cdot h^{f_{\varepsilon_j}}}, F_6 = g^{f_\eta}$, for each $i \in \{1, \dots, n\}$, $F_{7i} = g^{f_{\kappa_i}} h_i^{f_\eta}, F_{8i} = \prod_{i=1}^n v_i^{\alpha \cdot f_{\kappa_i} \cdot h_i^{f_{\varepsilon_i}}}$. Then, the $\mathcal{P}$ calculates $\varphi = H(pk, R, tg, F_1, F_2, F_3, F_4, F_{5i}, F_6, F_{7i}, F_{8i}, m)$. Finally, the $\mathcal{P}$ computes $\Gamma_1 = f_z + z \cdot \varphi, \Gamma_2 = f_\rho + \rho \cdot \varphi, \Gamma_3 = f_\eta + \eta \cdot \varphi, \Gamma_4 = f_w + w \cdot \varphi$, for each $i \in \{1, \dots, n\}, K_i = f_{\kappa_i} + \kappa_i \cdot \varphi, V_i = f_{v_i} + v_i \cdot \varphi$, and sends (m, σ) to the $\mathcal{V}$, where $\sigma = \{\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4, (K_i, V_i)_{i \in [1, n]}, \varphi, R, tg\}$.

The $\mathcal{V}$ receives message (m, σ) from $\mathcal{P}$. Then, $\mathcal{V}$ computes $F_1 = g^{\Gamma_1} / \left(R^\varphi \cdot \prod_{i=1}^t pk_i^{c \cdot K_i} \right), F_2 = g^{\Gamma_2} / ct_1^\varphi, F_3 = (pk_t^{\Gamma_2} \cdot g^{\Gamma_1}) / ct_2^\varphi, F_4 = g^{\Gamma_4} / T_0^\varphi, F_{5i} = \left(\prod_{i=1}^n g^{K_i} \cdot h^{\Gamma_4} \right) / T_1^\varphi,$

$F_6 = g^{\Gamma_3} / v_0^\varphi$, for each $i \in \{1, \dots, n\}, F_{7i} = (g^{K_i} \cdot h_i^{\Gamma_3}) / v_i^\varphi, F_{8i} = \left(\prod_{i=1}^n v_i^{\alpha \cdot K_i} \cdot h_i^{V_i} \right) / \left(\prod_{i=1}^n v_i^{\varphi \alpha i} \right)$. Then, the verifier $\mathcal{V}$ checks if $\varphi = H(pk, R, tg, F_1, F_2, F_3, F_4, F_{5i}, F_6, F_{7i}, F_{8i}, m)$.

Theorem 1: In the random oracle model, the presented Σ -protocol scheme is a non-interactive zero-knowledge argument. The following is the Σ -protocol security proof.

Completeness. In this study, direct verification follows to ensure completeness.

Soundness. We demonstrate that the Σ -protocol is soundness under the random oracle model, and the unforgeability of the Schnorr signature scheme depends on the DL assumption. Assume that an accepted argument (m, σ) is generated by a PPT prover $\mathcal{P}$, where $\sigma = \{\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4, (K_i, V_i)_{i \in [1, n]}, \varphi, R, tg\}$. Then, we construct a such an extractor EXT , which rewind $\mathcal{P}$ back to oracle query $H(pk, R, tg, F_1, F_2, F_3, F_4, F_{5i}, F_6, F_{7i}, F_{8i}, m)$ and return the result to φ . It then reprograms the $\varphi^* = H(pk, R, tg, F_1, F_2, F_3, F_4, F_{5i}, F_6, F_{7i}, F_{8i}, m)$ with $\varphi \neq \varphi^*$, and continue the remaining execution of prover $\mathcal{P}$. In expected PPT, obtain another valid argument (m, σ^*) , where $\sigma^* = \{\Gamma_1^*, \Gamma_2^*, \Gamma_3^*, \Gamma_4^*, (K_i^*, V_i^*)_{i \in [1, n]}, \varphi^*, R^*, tg^*\}$. Then, extractor EXT can computes as follows.

$$\begin{aligned} z &= \frac{\Gamma_1 - \Gamma_1^*}{\varphi - \varphi^*}, \quad \rho = \frac{\Gamma_2 - \Gamma_2^*}{\varphi - \varphi^*}, \quad \eta = \frac{\Gamma_3 - \Gamma_3^*}{\varphi - \varphi^*}, \\ w &= \frac{\Gamma_4 - \Gamma_4^*}{\varphi - \varphi^*}, \quad \kappa_i = \frac{K_i - K_i^*}{\varphi - \varphi^*}, \quad v_i = \frac{V_i - V_i^*}{\varphi - \varphi^*}, \end{aligned}$$

where $i \in [1, n]$, for the sake of brevity, we omit the modular operation of the above equations.

According to the aforementioned description, with a non-negligible possibility, it seems possible to effectively compromise the DL assumption. Therefore, the Σ -protocol satisfies the criterion of special soundness.

Special honest verifier zero knowledge. Without relying on the conventional Fiat-Shamir heuristic, by constructing simulator Sim a perfect zero-knowledge is established, which simulates any interaction with the verifier $\mathcal{V}$. The Sim randomly selects $(m, \sigma^* \in \mathbb{Z}_p^*)$, where $\sigma^* = \{\Gamma_1^*, \Gamma_2^*, \Gamma_3^*, \Gamma_4^*, (K_i^*, V_i^*)_{i \in [1, n]}, \varphi^*, R^*, tg^* \in \mathbb{Z}_p^*\}$ and computes $F_1^* = g^{\Gamma_1^*} / \left(R^{*\varphi^*} \cdot \prod_{i=1}^t pk_i^{c \cdot K_i^*} \right), F_2^* = g^{\Gamma_2^*} / ct_1^{\varphi^*}, F_3^* = (pk_t^{\Gamma_2^*} \cdot g^{\Gamma_1^*}) / ct_2^{\varphi^*}, F_4^* = g^{\Gamma_4^*} / T_0^{*\varphi^*}, F_{5i}^* = \left(\prod_{i=1}^n g^{K_i^*} \cdot h^{\Gamma_4^*} \right) / T_1^{*\varphi^*}, F_6^* = g^{\Gamma_3^*} / v_0^{*\varphi^*}$, for each $i \in \{1, \dots, n\}, F_{7i}^* = (g^{K_i^*} \cdot h_i^{\Gamma_3^*}) / v_i^{\varphi^*}, F_{8i}^* = (g^{K_i^*} \cdot h_i^{\Gamma_3^*}) / v_i^{\varphi^*}$.

Then, the simulator Sim sets the random oracle $\varphi^* = \mathcal{O}_H(pk, R^*, tg^*, F_1^*, F_2^*, F_3^*, F_4^*, F_{5i}^*, F_6^*, F_{7i}^*, F_{8i}^*, m)$, and Sim returns the result $(m, \{\Gamma_j^*\}_{j \in [1, 4]}, (K_i^*, V_i^*)_{i \in [1, n]}, tg^*, R^*, \varphi^*)$. Moreover, the $\{\Gamma_j^*\}_{j \in [1, 4]}, (K_i, V_i)_{i \in [1, n]}, \varphi^*, R, tg$ in the Sim possess a same distribution as the result of a transcript that is truthfully generated. Therefore, we demonstrate that the Σ -protocol adheres to the zero-knowledge property.

B. Security Proof

In this section, we prove that the BAST protocol can meet the security requirements of Definition 1 (e.g., Section III-H). We describe that the BAST satisfies unforgeability and

traceability, ensuring that the malicious signer ES can be traced by the blockchain or the adversary. That is, no PPT adversary $\mathcal{A}$ can distinguish valid forgery of the BAST. To arrive at a definition which is both formal and specific, we denote $(n, t, C) \leftarrow \mathcal{A}(\lambda)$, where $C \subseteq [n]$, t is threshold, $(pk, sk, sk_{ag}, sk_t) \leftarrow TKG(\lambda, n, t)$ as the trace key of ES generation algorithm, $\sigma_i \leftarrow \text{Sign}(sk_i, m, C)$ as the message share signature algorithm, $\sigma \leftarrow \text{RAG}(sk_{ag}, m, C, \{\sigma_i\}_{i \in C})$ as the RSU aggregate signature algorithm, $\{0, 1\} \leftarrow \text{Verify}(pk, m, \sigma)$ as the vehicle verification algorithm, and $C/\text{fail} \leftarrow \text{BTrace}(sk_t, m, \sigma)$ as the blockchain trace signature algorithm.

Lemma 1: This BAST protocol $\mathcal{Z} = (\text{Verify}, \text{BTrace}, \text{RAG}, \text{TKG}, \text{Sign})$ is considered traceable if for every adversary $\mathcal{A}$ that attacks BAST, there exists an adversary $\mathcal{B}$ that operates almost simultaneously with $\mathcal{A}$, such that

$$\text{Adv}_{\mathcal{A}, \text{BAST}}(\lambda) \leq (\text{Adv}_{\mathcal{N}}(\lambda)) \cdot q(\lambda) + \varepsilon(\lambda),$$

where the rigor of proof system and knowledge errors are denoted by q and ε .

Proof 1: The games following demonstrate this lemma.

Game 0. When $\mathcal{A}$ seeks assistance from challenger $\mathcal{B}$ with an oracle query, challenger $\mathcal{B}$ will respond, and the detailed process is as follows.

Send(Π_A, req). The $\mathcal{A}$ run $\mathcal{A}(\lambda)$, after receiving the query, the challenger $\mathcal{B}$ outputs n, t, C for $\mathcal{A}$, where t is the threshold number, and C is the number of participants.

Send(Π_{TA}, req). When the challenger $\mathcal{B}$ receives the query information, $\mathcal{B}$ randomly selected $sk_1, \dots, sk_n \in \mathbb{Z}_p^*$, calculates the corresponding public key $pk_i = g^{sk_i}, i \in [1, \dots, n]$. The $\mathcal{B}$ computes $T_0 = g^\omega$ and $T_1 = g^{t\omega}$ for encrypt t . Moreover, the $\mathcal{B}$ generates the random private key $sk_e, sk_{rs} \in \mathbb{Z}_p^*$, calculates the corresponding public key $pk_t = g^{sk_e}$ and $pk_{rs} = g^{sk_{rs}}$, respectively. Finally, the $\mathcal{B}$ generates the trace key $sk_t = \{pk, sk_e, t\}$, public key $pk = \{pk^*, pk_t, pk_{rs}, vk_{TA}, T_0, T_1\}$, and the combined key $sk_{ag} = \{pk, sk_{rs}, t, \omega\}$ for $\mathcal{A}$.

Send(Π_{ES}, m). When the challenger $\mathcal{B}$ receives the information, $\mathcal{B}$ selects randomly $R_i \in \mathbb{G}_1$, i.e., $R = \prod_{i \in C} R_i$. Then, the $\mathcal{B}$ computes $c = H(pk, R, m)$, chooses randomly $z_i \in \mathbb{Z}_p^*$, $g^{z_i} = pk_i^c \cdot R_i$, and $\sigma_i = (R_i, z_i)$. Finally, the $\mathcal{B}$ sends the signature information σ_i to the $\mathcal{A}$.

Send(Π_{RSU}, m). When the challenger $\mathcal{B}$ receives the information, $\mathcal{B}$ computes $R = \prod_{i \in C} R_i$, $z = \sum_{i \in C} z_i$, $c = H(pk, R, m)$. Then, the $\mathcal{B}$ computes $ct_1 = g^\rho, ct_2 = g^z pk_t^\rho$ through selects randomly $\rho \in \mathbb{Z}_p^*$, and calculates the $g^z = [\prod_{i=1}^t (pk_i)^{b_i}]^c \cdot R$, where $b_i = 1$ if $i \in C$. Then, the $\mathcal{B}$ uses zero-knowledge proof knowledge to generate the $\pi = \{z, \rho, \eta, \lambda, \kappa_1, \dots, \kappa_n, v_1, \dots, v_n\}$. Finally, the $\mathcal{B}$ sends the signature information $\sigma = \{R, ct_1, ct_2, \pi, tg, TS_{exp}\}$ to the $\mathcal{A}$, where TS_{exp} is a timestamp.

Send($\Pi_{vehicle}, m$). When the challenger $\mathcal{B}$ receives the information, $\mathcal{B}$ computes $c = H(pk, R, m)$, and valid of verification the $(pk_{rs}, (m, R, ct, \pi), tg)$, if the message is valid, the authenticity of message is demonstrated by the flag information that the challenger $\mathcal{B}$ returns to $\mathcal{A}$. Otherwise, $\mathcal{B}$ returns $\perp$ and rejects the communication.

Send($\Pi_{Blockchain}, m$). When the challenger $\mathcal{B}$ receives the information, $\mathcal{B}$ verifies the legitimacy of the message. If

the above verification is successful, the $\mathcal{B}$ chooses randomly $z^* \in \mathbb{Z}_p^*$ for computes $g^{z^*} = \frac{ct_2}{ct_1^{sk_e}}$. Then, the $\mathcal{B}$ calculates $pk_C = \prod_{i \in C} pk_i$, where $C \subseteq [n]$. Moreover, $\mathcal{B}$ calculation and verification whether the equation $g^{z^*} = R \cdot (pk_C)^c$ holds. If equal, the $\mathcal{B}$ return C to the $\mathcal{A}$, otherwise, the $\mathcal{B}$ outputs the fail to the $\mathcal{A}$.

In conclusion, Game 0 simulates a real attack. If W_0 is the event that $\mathcal{A}$ wins in Game 0, we have.

$$\text{Adv}_{\mathcal{A}, \text{BAST}}(\lambda) = \Pr[W_0].$$

Game 1. The $\mathcal{A}$ obtains the public key $pk = \{pk_1, \dots, pk_n, pk_t, pk_{rs}, vk_{TA}, T_0, T_1\}$ and the combined key $sk_{ag} = \{pk, t, sk_{rs}, \omega\}$. Game 1 is similar to Game 0, with the exception that Game 1 that the $\mathcal{A}$ must outputs a valid forgery message (m^*, σ^*) , where $\sigma^* = \{R^*, ct_1^*, ct_2^*, \pi^*, tg^*, TS_{exp}^*\}$.

We established adversary $\mathcal{A}^*$ for the Game 1, the $\mathcal{A}^*$ initiates a request to $\mathcal{A}$ until it obtains $\{pk_1, \dots, pk_n, pk_t, T_0, T_1, c^*, R^*, ct_1^*, ct_2^*\}$. Then, $\mathcal{A}^*$ run the extractor $\mathcal{A}^*$ for the proof system (P, V) , and outputs a witness $\psi = \{z^*, \rho^*, \eta^*, \kappa_1^*, \dots, \kappa_n^*, \lambda^*, v_1^*, \dots, v_n^*\}$. If ψ is a valid witness, $\mathcal{A}^*$ generates π^* and tg^* for computes σ^* . Then, $\mathcal{A}^*$ outputs (m^*, σ^*, ψ) .

According to the definition of extractor Ext , if W_1 is the event that $\mathcal{A}^*$ wins in Game 1, we have.

$$\Pr[W_1] \geq (\Pr[W_0] - \varepsilon(\lambda))/q(\lambda),$$

where the rigor of proof system and knowledge errors are denoted by q and ε .

Game 2. The differences between Game 1 and Game 2 are mostly because $\mathcal{N}$ has the same advantage over $\mathcal{A}^*$ in terms of its capacity to compromise the security of the underlying Schnorr signature. According to the above description, if W_1 is the event, adversary $\mathcal{N}$ wins Game 2, we have.

$$\text{Adv}_{\mathcal{N}}(\lambda) \geq \Pr[W_1].$$

Finally, by combining equation Game 0-2, we have.

$$\text{Adv}_{\mathcal{A}, \text{BAST}}(\lambda) \leq (\text{Adv}_{\mathcal{N}}(\lambda)) \cdot q(\lambda) + \varepsilon(\lambda).$$

C. Informal Theoretical Security Analysis

In this section, we show that the BAST protocol can meet the above security objectives (e.g., Section III-B). The detailed analysis process is as follows.

1) *Message Authentication:* Building on the adopted Σ -protocol flawless completeness and soundness, PPT adversaries are incapable of forging valid messages and signatures. Moreover, the message receiver can verify the authenticity of the message by inspecting $\text{Verify}(pk_{rs}, (m, R, ct, \pi), tg) = 1$ and $\text{Verify}(pk, m, \sigma) \rightarrow 1$ hold. The former is a proof used to verify that π is valid, and the latter is used to ensure the validity of messages and signatures.

2) *Malicious ES Traceability:* For the malicious signer ES, this proposed scheme provides a traceability function. The blockchain has the ability to find a $C \subseteq [n]$ and the size of c is equal to t , i.e., $g^z = R \cdot (\prod_{i \in C} pk_i)^c$. Therefore, the proposed scheme meets the malicious ES traceability.

3) *Resistance to 51% Attacks*: The proposed scheme might be compromised if a group of attackers controls more than half of the processing power. However, we implement a blockchain wherein the record nodes are trusted and resilient against malicious attacks.

4) *Resistance Collusion Attacks*: To prevent collusion between ES and RSU for the generation of valid messages and signatures. The vehicle verifies the message and signature received from the RSU. If the verification failure, the receipt of the message is rejected, and both the message and signature are reported to the blockchain. The blockchain will obtain the malicious ES of the signature by calculating the $g^z = R \cdot (\prod_{i \in C} pk_i)^c$. Moreover, we consider that multiple malicious ESs collude to generate malicious signatures, sending their individual signatures to the RSU to generation an aggregate signature. The vehicle verifies the validity of the signature by calculating $Verify(pk_{rs}, (m, R, ct, \pi), tg) = 1$, thereby preventing malicious signature information. Therefore, the above collusion attack is thwarted by the proposed scheme.

5) *Resistance Modification Attacks*: The request will be rejected if the vehicle or blockchain is unable to successfully validate the received message and signature.

6) *Resistance Impersonation Attacks of Vehicle*: The proposed scheme employs $Verify_{vk_{TA}}(Cert_{vid_i}, vid_i) \rightarrow true$ and $TS_B \leq TS_{vid_i}$ for legality verification of the vehicle identity before trace the signature of the messages reported by the vehicle to the blockchain.

7) *Resistance Replay Attacks*: In this proposed scheme, both the vehicle and the blockchain will reject replay requests messages that do not pass timestamp freshness verification.

D. Comparison of Security and Functionality

In this section, we conducted a security and functionality comparison of the proposed scheme with comparable schemes [3], [8], [35], and [47]. Since the analyses in [3], [8], [35], and [47], are similar concerning security and functionality. Here, we further analyze the security and functionality of [3]. The scheme [3] employs a (t, n) threshold signature to resist collusion attacks involving multiple ENs. Moreover, the ENs in the scheme are capable of directly providing services to vehicles. However, the scheme does not incorporate blockchain technology into the design of the vehicular network. A detailed analysis of other security properties is provided below.

1) *Support Mutual Authentication*: The scheme achieves mutual authentication between vehicles and ENs by verifying equations $e(Y, P) = e(R_1, P_{pub}) \cdot e(H_1(ID_V || TS || R_1), R_2)$ and $e(\sum_{k \in T} \sigma_k) = e(H(EXP), \sum_{k \in T} PK_{F_k})$. Only legitimate vehicles with secret key $sk_V = xH(ID_V)$ can generate valid $Y = r_1(sk_V + H_1(ID_V || TS || R_1))$. Similarly, only legitimate EN with the corresponding private key s_i can generate valid signature.

2) *Resistance to Replay Attacks*: During authentication, the vehicle transmits a message $\{ID_V, Y, R_1, R_2, TS\}$ to the EN. If the included timestamp TS has expired, the proof $\{Y, R_1, R_2\}$ cannot be authenticated. Therefore, the scheme effectively resistance replay attacks.

TABLE I
COMPARISON OF SECURITY AND FUNCTIONALITY

Description	[3]	[47]	[8]	[35]	Our
Support mutual authentication	✓	✓	✓	✓	✓
Resist collusion attacks	✓	×	✓	×	✓
Resist common attacks	✓	✓	✓	✓	✓
Support traceability for signer	×	✓	×	×	✓
Support blockchain integration with vehicular networks	×	✓	×	✓	✓
Get content directly from ES	✓	×	✓	✓	✓

3) *Resistance to Impersonation Attacks*: To impersonate a legitimate vehicle, an attacker must generate a valid proof $Y = r_1(sk_V + H_1(ID_V || TS || R_1))$. However, without access to the vehicle secret key sk_V , the attacker cannot construct a verifiable Y . Even if the attacker attempts to forge a value $Y' = r'_1(sk_V + H_1(ID_V || TS || R_1))$ based on the intercepted Y , the timestamp TS contained in Y' would be expired, and thus the proof would fail verification by the EN. Therefore, the scheme is capable of resisting vehicle impersonation attacks.

Table I provides a comparison of our scheme with schemes [3], [8], [35], [47] in terms of security and functionality. The results show that our scheme has greater advantages.

VI. PERFORMANCE EVALUATION

In this section, through a combination of theoretical analysis and experimental approaches, we implement a thorough assessment to determine the efficacy of the proposed scheme and the performance of related schemes [3], [8], [35], [47]. In this study, we implements $e : \mathbb{G}_1 \times \mathbb{G}_1 \rightarrow \mathbb{G}_T$ to achieve an 80-bit security level. Constructed by P over on the curve $E : y^2 = x^3 + x \mod p$, the $\mathbb{G}_1$ with order q is an additive group, where p and q are prime numbers with bit lengths of 160 bits and 512 bits, respectively. Furthermore, the ECC-based scheme usually employs the prime numbers $\bar{p}$ and $\bar{q}$. The additive group $\mathbb{G}$ on the non-singular elliptic curve $E : y^2 = x^3 + ax + b \mod \bar{p}$, where $\bar{p}, \bar{q}$ are 160 bits prime numbers, $a, b \in \mathbb{Z}_{\bar{p}}^*$.

A. Computational Cost Analysis

We let T_{ebp} , T_{bp} , T_{bpm} , T_{ecm} , T_{ap} , T_h , T_{sig} , T_{ver} , T_{AES} , T_B , and T_{MAC} denote the time needed to exponentiate operation over $\mathbb{G}_T$, a bilinear pairing operation, a scale multiplication operation over $\mathbb{G}_1$, a scale multiplication operation over $\mathbb{G}$, a point addition operation over $\mathbb{G}_1$, a MapToPoint hash operation over $\mathbb{Z}_p^*$, a message signature operation, a message verification operation, a symmetric encryption/decryption operation using AES algorithm in cipher block chaining mode, a Bloom filter lookup operation, and a message authentication code (MAC) operation, respectively. Note, this scheme ignores for the time required to perform XOR and $\parallel$ operations.

To facilitate the computation of the computational overhead of [3], [8], [35], and [47], and the corresponding entities in the proposed scheme, We utilizes the JPBC¹ cryptography library to measure the time taken for cryptographic operations across various experimental platforms. We simulate the estimated time required for vehicle, ES/RSU/EN, and TA/ASP to

¹<http://gas.dia.unisa.it/projects/jpbc/>

TABLE II
COMPUTATIONAL COST COMPARISON AMONG THE PROPOSED SCHEME AND RELATED SCHEMES

Schemes	Signature Generation	Signature Verification
Yang et al. [3]	$(8 + 2t)T_{ecm} + (t + 14)T_h + (t + 9)T_{bp} + (4t + 1)T_{ap}$	$(4t + 3)T_{bp} + (5t + 3)T_h + 2tT_{ecm} + (t + 1)T_{bpm}$
Li et al. [47]	$(8(n - t) + 3)T_{ecm} + 3(n - t)T_{ap} + ((n - t) * t + 1)T_{AES} + (2n + 2)T_h$	$4nT_{ecm} + 2rT_{ap} + (n - t)T_{AES} + (2n + 2)T_h$
Baza et al. [8]	$2T_{sig} + t(t - 1)T_{ebp} + 2(t - 1)T_{bp} + (t + 1)T_h + T_{ver} + 2tT_{bpm} + (t - 1)T_{ap}$	$2T_{ap} + tT_{ebp} + tT_h + T_{ver}$
Li et al. [35]	$(t + 1)T_{MAC} + tT_{ebp} + (3t + 1)T_h + (1 + 5t)T_{bpm} + (3t - 1)T_{ap} + 2tT_{bp}$	$T_B + 2T_{bp} + T_{ebp} + (\lceil \log_2^n \rceil + 1)T_h$
Our scheme	$(4n + t + 4)T_{ebp} + (t + 1)T_h + (4t + 2)T_{ecm} + (n + 2t)T_{ap} + T_{sig}$	$T_h + T_{ver} + (3n + t + 4)T_{ebp} + (2n + 3)T_{ecm}$

TABLE III
EXECUTION TIME OF BASIC OPERATIONS (MS)

Symbol	$Time_1$ (ms)	$Time_2$ (ms)	$Time_3$ (ms)
T_{ebp}	11.7227	1.1492	2.0267
T_{bp}	52.8025	4.8348	9.6244
T_{bpm}	68.1495	6.5066	13.1674
T_{ecm}	39.6402	3.8791	7.7675
T_{ap}	17.4778	1.6963	3.6892
T_h	0.0108	0.0008	0.0015
T_{sig}	13.2282	1.193	2.4414
T_{ver}	26.0281	2.3933	4.6805
T_{AES}	0.319	0.0792	0.0302
T_B	0.0146	0.00271	0.00756
T_{MAC}	0.8023	0.0705	0.1322

perform cryptographic operations, employing Raspberry Pi 3 model B, common desktop computer, and cloud server (CS) (ecs.t6-c1m1.large, 2 vCPU, 2 GiB, Ubuntu 18.04, 64 bits). As indicated in Table III, the vehicle, ES, and CS execution times are represented by the variables $Time_1$, $Time_2$, and $Time_3$, respectively.

A thorough study was conducted out to enable a thorough knowledge of the computational and communication overheads of [3], [8], [35], and [47], and the proposed scheme. Because the analysis of the [3], [8], [47], [35] is similar, we will focus on a detailed analysis of the Yang et al. [3] and our scheme. According to Table II, our theoretical analysis provides valuable insights into the performance of signature generation and verification in the relevant schemes. In this table, we let t represents the threshold of the threshold signature or ring signature, and n represents the size of the member set in the threshold signature.

Here, we conducted a detailed analysis of the computational overhead of Yang et al. [3] in the signature generation and signature verification processes. In [3], the process of signature generation and verification is separated into two phases. In the first phase, the vehicle performs signature verification in collaboration with a consortium of ENs, which collectively generate a valid token, the vehicle needs to perform $(5 + 2t)$ scale multiplication operations over $\mathbb{G}$, $(t + 11)$ hash operations, $(t + 9)$ bilinear pairing operations, and $(4t + 1)$ point addition operations, therefore, the computational cost for the vehicle at this phase is $(5 + 2t)T_{ecm} + (t + 11)T_h + (t + 9)T_{bp} + (4t + 1)T_{ap}$. And this EN needs to perform $4t$ bilinear pairing operations, $(5t + 1)$ hash operations, t scale multiplication operation over $\mathbb{G}_1$, and $2t$ point multiplication operations for the vehicle generates a valid token, therefore, the computational cost for the EN at this phase is $4tT_{bp} + (5t + 1)T_h + 2tT_{ecm} + tT_{bpm}$.

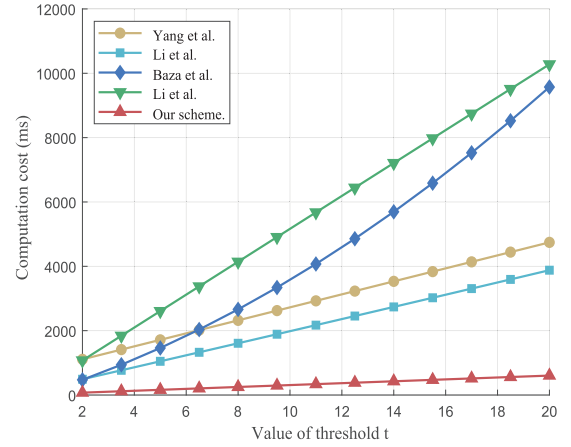


Fig. 4. Computational cost comparison for signature generation.

In the second phase, the vehicle needs to perform three scale multiplication operations over $\mathbb{G}$ and three hash operations. Therefore, the total computational overhead of the vehicle is $(8 + 2t)T_{ecm} + (t + 14)T_h + (t + 9)T_{bp} + (4t + 1)T_{ap}$. And this EN needs to perform two hash operations, three bilinear pairing operations, and one point multiplication operation over $\mathbb{G}_1$ to verify the signature. Therefore, the total computational cost of the EN is $(4t + 3)T_{bp} + (5t + 3)T_h + 2tT_{ecm} + (t + 1)T_{bpm}$.

In our scheme, the signature generation is composed of two phases, to enhance the analytical rigor, we conducted an in-depth examination of both phase in this scheme. In the first phase, to generate a shared signature, ES needs to perform t exponentiate operations, t hash operations, t point addition operations, and t scale multiplication operations over $\mathbb{G}_1$, therefore, the computational cost for the ES at this phase is $t(T_{ebp} + T_h + T_{ecm} + T_{ap})$. In the second phase, the RSU receives the signature sent by ES and needs to perform $(4n + 4)$ exponentiate operations, one hash operation, $(3t + 2)$ scale multiplication operations over $\mathbb{G}_1$, $(n + t)$ point addition operations, and one message signature operation. Therefore, the total computational overhead during signature generation is $(4n + t + 4)T_{ebp} + (t + 1)T_h + (4t + 2)T_{ecm} + (n + 2t)T_{ap} + T_{sig}$. In the signature verification phase, when the vehicle receives the aggregate signature sent by RSU, it needs to perform a hash operation, a message verification operation, $(3n + t + 4)$ exponentiate operations, and $(2n + 3)$ scale multiplication operations over $\mathbb{G}_1$ for verify the signature and message credibility. Therefore, the total computational overhead during signature verification is $T_h + T_{ver} + (3n + t + 4)T_{ebp} + (2n + 3)T_{ecm}$.

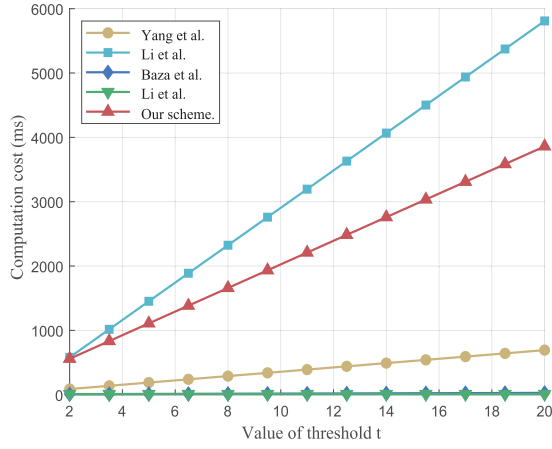


Fig. 5. Computational cost comparison for signature verification.

According to [35], to present a more complex and practical threshold signature environment, we let $n = 1.5t$, and the value of threshold t changes from 2 to 20. To visually compare the results, Fig. 4 and Fig. 5 show the computational overhead of the relevant schemes in signature generation and signature verification, respectively. Analyzing Fig. 4 reveals that the signature generation overhead of the proposed scheme is the lowest. Specifically, compared with [3], [8], [35], and [47], when the threshold $t = 20$, it is reduced compared to the total computational overhead of related schemes are about $(4744.46 - 603.522)/4744.46 = 87.28\%$, $(3879.45 - 603.522)/3879.45 = 84.44\%$, $(9571.89 - 603.522)/9571.89 = 93.69\%$, $(10278.4 - 603.522)/10278.4 = 94.13\%$, respectively.

Analyzing Fig. 5 illustrates that this schemes has a greater computational overhead for signature verification than related schemes. This is primarily due to the fact that the signature verification in the proposed scheme is performed by the vehicle. However, when both schemes utilize vehicles for signature verification, the proposed scheme demonstrates lower computational overhead compared to Baza et al. [8]. Specifically, when the threshold $t = 20$, the proposed scheme reduce computational overhead is about $(5809.35 - 3859.76)/5809.35 = 33.56\%$. Moreover, in this study, comparing the total overhead of signature generation and signature verification with related schemes, this scheme has lower computational overhead. Specifically, compared with [3], [8], [35], and [47], the decrease in overall computational cost are about 17.92%, 53.51%, 53.93%, 56.62%, respectively.

B. Communication Cost Analysis

According to the existing scheme, p is 64 bytes and $\bar{p}$ is 20 bytes, the elements $\mathbb{G}_1$ and $\mathbb{G}$ are represented using 128 bytes, and 40 bytes, respectively. Moreover, the size of the timestamp is set to 4 bytes and represented by T . And we consider the random number, output of the MAC function, general hash function, symmetric encryption/decryption, and message are set to 20 bytes and represented by $\mathbb{Z}_p^*$.

Since the analyses in [3], [8], [35], and [47], are similar concerning communication overhead, our detailed analysis delves into the calculation process of communication overhead

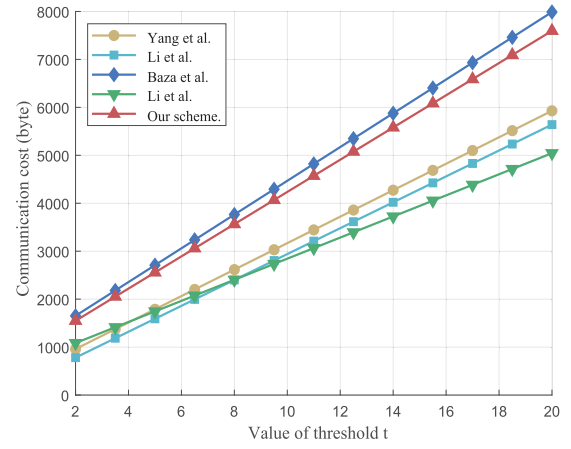


Fig. 6. Communication cost comparison for signature generation.

in both Yang et al. [3] and the proposed scheme. In [3], we primarily analyze the communication overhead in the signature (token) generation process. The vehicle sends a request $req = (ID_V, Y = r_1(sk_V + H_1(ID_V||TS||R_1), R_1 = r_1H(ID_V), R_2 = r_1P, TS)$, and at least t ENs sends the message $\{V_k = r_2^k P, W_k = \sigma_k \oplus H_2(U_k), C = EXP \oplus H_3(U_k)\}$. Therefore, the total ciphertext size required to generate a signature is $(2t + 3)|\mathbb{G}_1| + (t + 1)|\mathbb{Z}_p^*| + |T|$.

In this proposed scheme, generating a complete signature involves two parts (i.e., the ES shared signature and the aggregate signature generated by RSU). Therefore, when calculating the communication overhead, the sum of these two parts must be considered. These t ESs individually send their signature information $\sigma_i = \{R_i = g^{r_i}, z_i = r_i + sk_i \cdot c_i\}$ to RSU, requesting the generation of a complete signature, and the RSU sends this full signature $\sigma = \{R = \prod_{i \in C} R_i, ct_1 = g^p, ct_2 = g^z p k_t^p, \pi, tg, TS_{exp}\}$. Therefore, the total ciphertext size required to generate a signature is $(2t + 6)|\mathbb{G}_1| + (2n + t + 5)|\mathbb{Z}_p^*| + 2|T|$.

Based on the above theoretical analysis, we can determine the communication overhead during the signature generation process. As shown in Fig. 6, the comparison results of communication overhead between the proposed scheme and related schemes are clearly visible. For [3], [8], [35], [47], and our scheme, when the threshold $t = 20$, the total communication overhead is 5928 bytes, 5640 bytes, 7988 bytes, 5044 bytes, and 7596 bytes, respectively. Despite the notably higher communication overhead observed in this scheme compared to [3], [35], and [47], this discrepancy is primarily attributed to the signature generation process, which necessitates RSU aggregation. It should be noted that in practical scenarios where RSU and ES are connected through wires network, the actual communication overhead is expected to be lower than what is indicated by theoretical analysis.

C. Blockchain-Related Costs Analysis

The proposed scheme is implemented on the Ethereum test network Goerli² for evaluate its feasibility for blockchain

²<https://goerli.etherscan.io/>

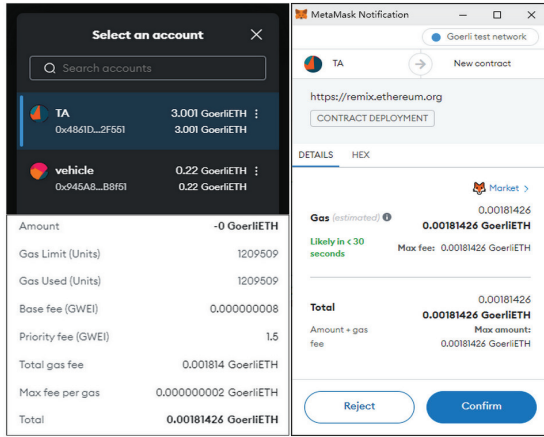


Fig. 7. The deployment of smart contract.

implementation. Here, Goerli provides free fund requests for the implementation of smart contracts and features a user-friendly interface for block browsing. We connect the Remix³ compiler in Chrome and the Ethereum network Rinkeby employing the MetaMask⁴ plug-in in Google Chrome for deploy and run the developed smart contract. Notably, the Goerli network represents any blockchain based on the Ethereum protocol, whether it is a public or private network. Additionally, we do not consider the overhead in private networks. The specific deployment and testing processes as follows.

First and foremost, we utilize MetaMask to create two accounts (TA, vehicle) for evaluating this scheme, designated as addresses $0 \times 4861D85334EfB0c4cdD11a19E33BAcB759b2F551$, and $0 \times 945A81106aD17654F075111469aC89b9A00B8f51$. Then, switch to the TA account and request a certain amount of Ethers from Goerli to load the private key trace the malicious signature into the blockchain. In this way, the TA account is capable of executing various operations, including vehicle identity verification, malicious signature trace, publishing malicious vehicle information.

Besides, we perform the following operations as TA. As shown in Fig. 7, we use Remix to deploy the smart contract to the Goerli and obtain its address (i.e., $0 \times 5ae99bb5a1386b660a7c195babf15f20ba89816b$). Moreover, we invoke the vehicle request algorithm using Remix and add the vehicle request information to the Goerli, to assess the overhead in transaction fees, we evaluated related operations, including vehicle authentication, malicious signature trace, and publishing malicious vehicle or signer information.

Last but not least, according to the data presented in Table IV, the gas cost associated with the execution of the pertinent operations has been determined. Specifically, the exchange rate is 1 ETHER = 2240.42 USD, the gas maximum is 10,000,000 gas, and the average price per GAS is 1.5 GWEI. According to Table IV, the deployment (i.e., Deploy), the process with the highest cost associated with smart contracts, costs about 4.064712255, but it only executes once. Moreover,

TABLE IV
THE SMART CONTRACT OPERATION GAS COST

Operation	Gas Used	Actual Cost (ether)	USD
Deploy	1209509	0.001814263510885581	4.064712255
Vehicle request	32337	0.000048505500226359	0.108672693
Vehicle authentication	31575	0.000047362500284175	0.106111893
Malicious signature trace	44718	0.000067077000357744	0.150280653
Publishing malicious vehicle	24069	0.000036103500168483	0.080887004
Publishing malicious signer	24043	0.000036064500192344	0.080799628

the cost for other operations is below 0.16, making it an acceptable cost for blockchain transactions even with frequent calls to these operations.

VII. CONCLUSION

Herein, we proposed a blockchain-assisted secure and traceable data sharing scheme for cache-based vehicular networks. This scheme addressed the reliability of ES broadcast data and achieved traceability of a malicious signer ES. Moreover, the BAST enhanced the efficiency of vehicle verification for ES broadcast data using non-interactive signatures of knowledge based on the Σ -protocol. Only a single interaction with the ES was required to obtain the necessary data, which significantly improved the practicality of the scheme. According to the security analysis, this scheme successfully met the vehicular networks security objectives. In terms of performance evaluation, our scheme demonstrated greater acceptability to blockchain-assisted vehicular networks, specifically regarding signature generation and traceability.

However, this scheme still faces challenges in improving the efficiency of vehicle verification, particularly due to limitations in vehicle computing resources and network conditions. Future research will focus on optimizing the vehicle verification process with lightweight encryption algorithms, edge computing, and network optimization. For example, outsourcing vehicle computing tasks to edge servers can further reduce verification overhead on the vehicle side. The edge server will be responsible for performing complex encryption and verification tasks, while the vehicle side will handle only lightweight computing tasks. This approach will greatly improve system efficiency and reduce the burden on the vehicle side.

VIII. ACKNOWLEDGMENT

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³<https://remix.ethereum.org/>

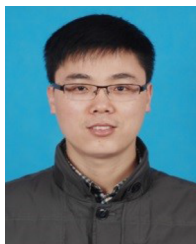
⁴chrome.google.com/detail/nkbihfheogaeaoehlfknodbefgpgknn

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